26900247 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEFTQ304 - POST HARVEST TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Which type of heating occurs during RF processing? [K] / [CO2]
2. Define microwave heating. [K] / [CO2]
3. Discuss two applications of supercritical fluid extraction in the spice processing industry? [U] / [CO4]
4. Define nanosensor. [K] / [CO5]
5. List out the factors which affecting Post harvest loss [An] / [CO1]
6. Explain any four areas included in post-harvest management. [U] / [CO1]
7. State two factors affecting membrane flux. [K] / [CO4]
8. Define nanotechnology in the context of the food industry. [U] / [CO5]

9. Justify the use of plant growth regulators as a method for extending the storage life of fruits and vegetables. [E] / [CO1]

10. What does PEF stand for? [K] / [CO2]

11. Explain any two important properties of Supercritical fluid extraction? [U] / [CO4]

12. Recognise one application of nanoparticles in food preservation. [K] / [CO5]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

13. Explain the importance of proper handling, storage and transportation in maintaining quality. [An] / [CO1]

14. Relate the importance of raw material selection to the quality of processed fruits and vegetables. [U] / [CO1]

15. Discuss the principle, advantages and applications of Supercritical Fluid extraction in food processing industry? [U] / [CO4]

16. Explain the significance of nanoparticles in the food processing sector. [U] / [CO5]

17. Explain the different operations involved in post-harvest management. [U] / [CO1]

18. Suggest a suitable application of inductive heating in the food industry and explain how the technology can improve processing efficiency. [A] / [CO2]

19. Differentiate between microfiltration and ultrafiltration based on membrane pore size, retained components and applications. [U] / [CO4]

20. Apply nanotechnology concepts to develop a suitable approach for food packaging and preservation. [A] / [CO5]

[5x6 = 30]

Part C

Long Essay Questions

Answer any **2** questions

Each question carries **10** marks

21. Debate the importance of proper irrigation, fertilization, pest management, and other pre-harvest practices in maintaining post-harvest quality and reducing losses. [An] / [CO1]

22. Explain the role of electrical conductivity of food materials in ohmic heating and discuss the factors affecting conductivity. [U] / [CO2]

23. Evaluate the suitability of reverse osmosis for fruit juice concentration with respect to product quality and processing efficiency. [E] / [CO4]

24. Propose a suitable nanocomposite-based packaging application for a selected food product and explain its expected benefits. [C] / [CO5]

[10x2 = 20]